



NAME:-

USAMA ARSHAD

REG ID:-

247141

DEPARTMENT:-

BS-CYBER SECURITY-FALL-24 (A)

SUBJECT:-

PROGRAMMING FUNDAMENTALS (Lab)

PRESENTED BY

USAMA ARSHAD

PROVIDED TO

MR. WASEEM YOUNAS

LAB # 07**Lab 7.1 A****Code:**

```
Lab 7.1 A.cpp
1  #include <iostream>
2  using namespace std;
3
4  // USAMA ARSHAD
5
6  #include <iostream>
7  using namespace std;
8  int main( )
9  {
10     // Declaring two variables
11     int num1, num2 ;
12
13     // Taking integers from user
14     cout << "Enter two integers to check equality " << endl;
15     cin >> num1 >> num2 ;
16
17     cout << "num1 = " << num1 << " and num2 = " << num2 << endl;
18
19     // Checking the equality
20     if (num1 == num2)
21     {
22
23         cout << endl << "The Values are same " << endl ;
24         cout << "Hey, that's a coincidence!" << endl;
25     }
26     // Handling the inequality
27     else
28     {
29         cout << endl << "The values are not the same" << endl;
30     }
31     return 0;
32 }
```

Output 1:

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 7\Lab 7.1 A.exe
Enter two integers to check equality
5
7
num1 = 5 and num2 = 7

The values are not the same

-----
Process exited after 12.03 seconds with return value 0
Press any key to continue . . .
```

Output 2:

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 7\Lab 7.1 A.exe
Enter two integers to check equality
7
7
num1 = 7 and num2 = 7

The Values are same
Hey, that's a coincidence!

-----
Process exited after 3.129 seconds with return value 0
Press any key to continue . . .
```

Lab 7 A – Task 1

Code:

```
1  #include <iostream>
2  using namespace std;
3  int main ()
4  {
5
6      // Deaclaration of variables
7      float bill_1, bill_2, bill_3, bill_4, bill_avg ;
8
9      // Taking Bill from user
10     cout << "Enter your Bill for 1st Quarter " << endl;
11     cin >> bill_1 ;
12     cout << "Enter Your Bill for 2nd Quarter " << endl;
13     cin >> bill_2 ;
14     cout << "Enter your Bill for 3rd Quarter " << endl;
15     cin >> bill_3 ;
16     cout << "Enter your Bill for 4th Quarter " << endl;
17     cin >> bill_4 ;
18
19     // Calculating average of Bills
20     bill_avg = ( bill_1 + bill_2 + bill_3 + bill_4 ) / 4 ;
21     cout << "The Average of all Bills = " << bill_avg << endl;
22
23     // Deploying Conditions for Water usage
24     if ( bill_avg >= 75 )
25     {
26         cout << "You are using excessive Amount of Water " << endl;
27     }
28     else if ( bill_avg >= 25 && bill_avg < 75 )
29     {
30         cout << "You are using typical Amount of Water " << endl;
31     }
32     else
33     {
34         cout << "We appreciate you Conserve Water" << endl;
35     }
36     return 0 ;
37 }
```

Output:

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 7\Lab 7A - Task 1.exe
Enter your Bill for 1st Quarter
150
Enter Your Bill for 2nd Quarter
360
Enter your Bill for 3rd Quarter
120
Enter your Bill for 4th Quarter
90
The Average of all Bills = 60
You are using typical Amount of Water

-----
Process exited after 18.89 seconds with return value 0
Press any key to continue . . .
```

Lab 7.2 A**Code:**

```
1
2 // USAMA ARSHAD
3
4 #include <iostream>
5 using namespace std;
6
7 int main()
8 {
9
10     float average; // holds the grade average
11
12     // Taking average from user
13     cout << "Input your average:" << endl;
14     cin >> average;
15
16     // Showing category accprding to average
17     if ( average > 100)
18         cout << "Enter Valid Average ( 1 - 100 ) " << endl ;
19
20     else if ( average >= 90)
21         cout << "You are in A category " << endl ;
22
23     else if ( average >= 80)
24         cout << "You are in B category " << endl ;
25
26     else if ( average >= 70)
27         cout << "You are in C category " << endl ;
28
29     else if (average >= 60)
30         cout << "You Pass" << endl;
31
32     else
33         cout << "You Fail" << endl;
34
35     return 0;
36 }
```

Output 1:

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 7\Lab 7.2 A.exe
Input your average:
89
You are in B category

-----
Process exited after 3.66 seconds with return value 0
Press any key to continue . . .
```

Output 2:

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 7\Lab 7.2 A.exe
Input your average:
58
You Fail

-----
Process exited after 4.156 seconds with return value 0
Press any key to continue . . .
```

Lab 7 A – Task 3

Code:

```
1  #include <iostream>
2  using namespace std;
3  int main ()
4  {
5      // declaring variables
6      int pay, exc_amnt1, exc_amnt2, exc_amnt3, exc_amnt4, exc_amnt5, exc_amnt6 ;
7      float tax ;
8
9      // Taking Annual salary from user
10     cout << "Enter your annual salary " << endl ;
11     cin >> pay ;
12
13     // Calculation of Tax
14     if ( pay >= 600000 && pay < 1200000 )
15     {
16         exc_amnt1 = pay - 600000 ;
17         tax = exc_amnt1 * ( 2.5 / 100 ) ;
18     }
19     else if ( pay >= 1200000 && pay <= 2400000 )
20     {
21         exc_amnt2 = pay - 1200000 ;
22         tax = ( exc_amnt2 * 12.5 / 100 ) + 15000 ;
23     }
24     else if ( pay >= 2400000 && pay <= 3600000 )
25     {
26         exc_amnt3 = pay - 2400000 ;
27         tax = ( exc_amnt3 * 20 / 100 ) + 165000 ;
28     }
29     else if ( pay >= 3600000 && pay <= 6000000 )
30     {
31         exc_amnt4 = pay - 3600000 ;
32         tax = ( exc_amnt4 * 25 / 100 ) + 405000 ;
33     }
34     else if ( pay >= 6000000 && pay <= 12000000 )
35     {
36         exc_amnt5 = pay - 6000000 ;
37         tax = ( exc_amnt5 * 32.5 / 100 ) + 1005000 ;
38     }
39     else if ( pay >= 12000000 )
40     {
41         exc_amnt6 = pay - 12000000 ;
42         tax = ( exc_amnt6 * 35 / 100 ) + 2955000 ;
43     }
44     else
45     {
46         cout << "ENJOY ! You are not eligible to pay Tax " << endl ;
47     }
48     cout << endl << "Your Annual Income tax is " << tax << endl ;
49     return 0 ;
50 }
```

Output 1:

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 7\Lab7A-Task3 M-Q3.exe
Enter your annual salary
450000
ENJOY ! You are not eligible to pay Tax

Your Annual Income tax is 0

-----
Process exited after 4.381 seconds with return value 0
Press any key to continue . . .
```

Output 2:

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 7\Lab7A-Task3 M-Q3.exe
Enter your annual salary
5700000
Your Annual Income tax is 930000

-----
Process exited after 6.218 seconds with return value 0
Press any key to continue . . .
```

Lab 7 B – Task 1**Code:**

```
1 // USAMA ARSHAD
2
3
4 #include <iostream>
5 using namespace std;
6
7 int main()
8 {
9     char year;
10    float gpa;
11
12    cout << "In which year you are studying ?" << endl;
13    cout << endl << "Enter 1 (freshman), 2 (sophomore),"
14    "3 (junior), or 4 (senior) " << endl << endl;
15    cin >> year;
16
17    cout << endl << "Now enter your GPA" << endl;
18    cin >> gpa;
19
20    if ( gpa >= 2.0 && year == '4')
21        cout << endl << "It is time to graduate soon" << endl;
22
23    else
24        cout << endl << "You need more schooling " << endl;
25
26    return 0;
27 }
```

Output 1:

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 7\Lab 7B - Task 1.exe
In which year you are studying ?

Enter 1 (freshman), 2 (sophomore),3 (junior), or 4 (senior)

4

Now enter your GPA
3.9

It is time to graduate soon

-----
Process exited after 6.597 seconds with return value 0
Press any key to continue . . .
```

Output 2:

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 7\Lab 7B - Task 1.exe
In which year you are studying ?

Enter 1 (freshman), 2 (sophomore),3 (junior), or 4 (senior)

1

Now enter your GPA
3.5

You need more schooling

-----
Process exited after 8.195 seconds with return value 0
Press any key to continue . . .
```

Lab 7 C – Task 1

Code:

```
2 // USAMA ARSHAD
3
4 #include <iostream>
5 using namespace std;
6 int main()
7 {
8     char grade;
9
10    // Taking Grade from user
11    cout << "What grade did you earn in PF ?" << endl;
12    cin >> grade;
13
14    // Appreciation for Grade Obtained
15    if (grade == 'A')
16    {
17        cout << "an A - excellent work !" << endl;
18    }
19    else if (grade == 'B')
20    {
21        cout << "you got a B - good job" << endl;
22    }
23    else if (grade == 'C')
24    {
25        cout << "You got C - satisfactory " << endl;
26    }
27    else if (grade == 'D')
28    {
29        cout << "There is Problem, D is passing " << endl;
30    }
31    else if (grade == 'F')
32    {
33        cout << "You are FAILED ! Better Luck Next Time " << endl;
34    }
35    else
36    {
37        cout << "Enter Correct Grade (A, B, C, D, F) " << endl;
38    }
39    return 0 ;
40 }
```

Output 1:

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 7\Lab 7C - Task 1.exe
What grade did you earn in PF ?
B
you got a B - good job

-----
Process exited after 10.33 seconds with return value 0
Press any key to continue . . .
```

Output 2:

```
D:\University\Programming Fundamentals (Lab)\Lab Tasks\Lab 7\Lab 7C - Task 1.exe
What grade did you earn in PF ?
b
Enter Correct Grade (A, B, C, D, F)

-----
Process exited after 3.795 seconds with return value 0
Press any key to continue . . .
```

Thank you for your time!

THE END